EEE301: Signals and Systems

Tutorial 6: Analysis of LTI Systems in *z*-Domain

1. Determine the *Z* transform of the following signals using the *z*-transform table.

   (a) $x[n] = (0.95)^n u[n]$

   (b) $x[n] = (1+n)u[n]$

   (c) $x[n] = \left(\frac{1}{2}\right)^n u[n] + \left(\frac{1}{3}\right)^n u[n]$

2. Using the z-transform table and z-transform properties, find z-transform of

   a) $2^{n+1}u[n-1] + e^{n-1}u[n]$
   b) $n\gamma^n u[n-1]$

3. Find the z-transform and the associated ROC for each of the following sequences:

   (a) $x[n] = \delta[n-n_0]$
   (b) $x[n] = u[n-n_0]$
   (c) $x[n] = a^{n+1}u[n+1]$
   (d) $x[n] = u[-n]$
   (e) $x[n] = a^{-n}u[-n]$

4. Determine the inverse *z* transform of

   (a) $X(z) = \dfrac{1+2z^{-1}}{1-2z^{-1}+z^{-2}}$ for $|z| > 1$

   (b) $X(z) = \dfrac{1+3z^{-1}}{1+3z^{-1}+2z^{-2}}$ for $|z| > 1$

5. Find the unit impulse of LTI discrete system with transfer functions

   a) $H(z) = \dfrac{z(z-4)}{z^2-5z+6}$ b) $H(z) = \dfrac{z(-5z+22)}{(z+1)(z-2)^2}$

   c) $H(z) = \dfrac{z-4}{(z-2)(z-3)}$

6. An LTI system is described by

   $$2y[n+2] - 3y[n+1] + y[n] = 4f[n+2] - 3f[n+1].$$

   a) Find the output $y[n]$ if initial conditions are $y(-1) = 0,\ y(-2) = 1$ and input is $f[n] = (4)^{-n}u[n]$ using z-transform.

b) Write out the transfer function of the system.

7. A causal discrete-time LTI system is described by

$$y[n] - \frac{3}{4}y[n-1] + \frac{1}{8}y[n-2] = x[n]$$

where $x[n]$ and $y[n]$ are the input and output of the system, respectively.

(a) Determine the system transfer function $H(z)$.
(b) Find the impulse response $h[n]$ of the system.
(c) Find the response of the system to unit step input $x[n] = u[n]$.

8. Show the realization for the system with transfer function

$$H(z) = \frac{z(3z-1.8)}{z^2 - z + 0.16}.$$

Appendix 1

Basic z-Transforms

Signal	Transform	ROC
$\delta[n]$	1	All z
$u[n]$	$\frac{z}{z-1}$	$\|z\|>1$
$nu[n]$	$\frac{z}{(z-1)^2}$	$\|z\| > 1$
$n^2u[n]$	$\frac{z(z+1)}{(z-1)^3}$	$\|z\| > 1$
$a^nu[n]$	$\frac{z}{z-a}$	$\|z\|>\|a\|$
$a^{n-1}u[n-1]$	$\frac{1}{z-a}$	$\|z\| > \|a\|$
$na^n u[n]$	$\frac{az}{(z-a)^2}$	$\|z\| > \|a\|$
$n^2a^n u[n]$	$\frac{az(z+a)}{(z-a)^3}$	$\|z\| > \|a\|$

Appendix 2

z-Transform Properties

$x(n)$	$X(z)$
$x(n-m_0)$	$z^{-m}\,X(z)$
$x_1(n) * x_2(n)$	$X_1(z)X_2(z)$
$x_1(n)x_2(n)$	$\frac{1}{2\pi}\oint X_1(u)X_2\left[\frac{z}{u}\right]u^{-1}du$
$x(-n)$	$X\left(\frac{1}{z}\right)$
$a^n\,x(n)$	$X\left(\frac{z}{a}\right)$
$n\,x(n)$	$-z\frac{d}{dz}X(z)$
$f(n-m)$ with initial conditions	$\frac{1}{z^m}X(z)+z^{-m}\sum_{k=1}^{m}x(-k)z^k$